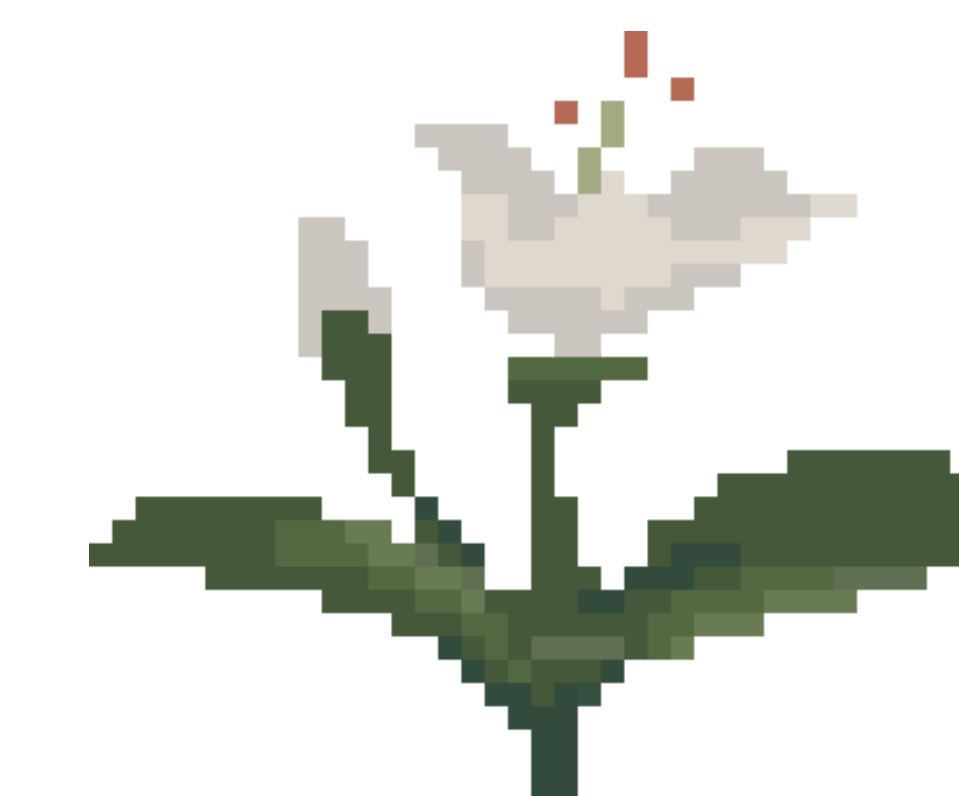


Exploring the Slime Mold Algorithm: AD NATURA



Motivation

We wish to bring a new experience to demonstrate the beauty of the non-human. We present the slime mold, an organism that defies all means to categorize and explain its queer properties.

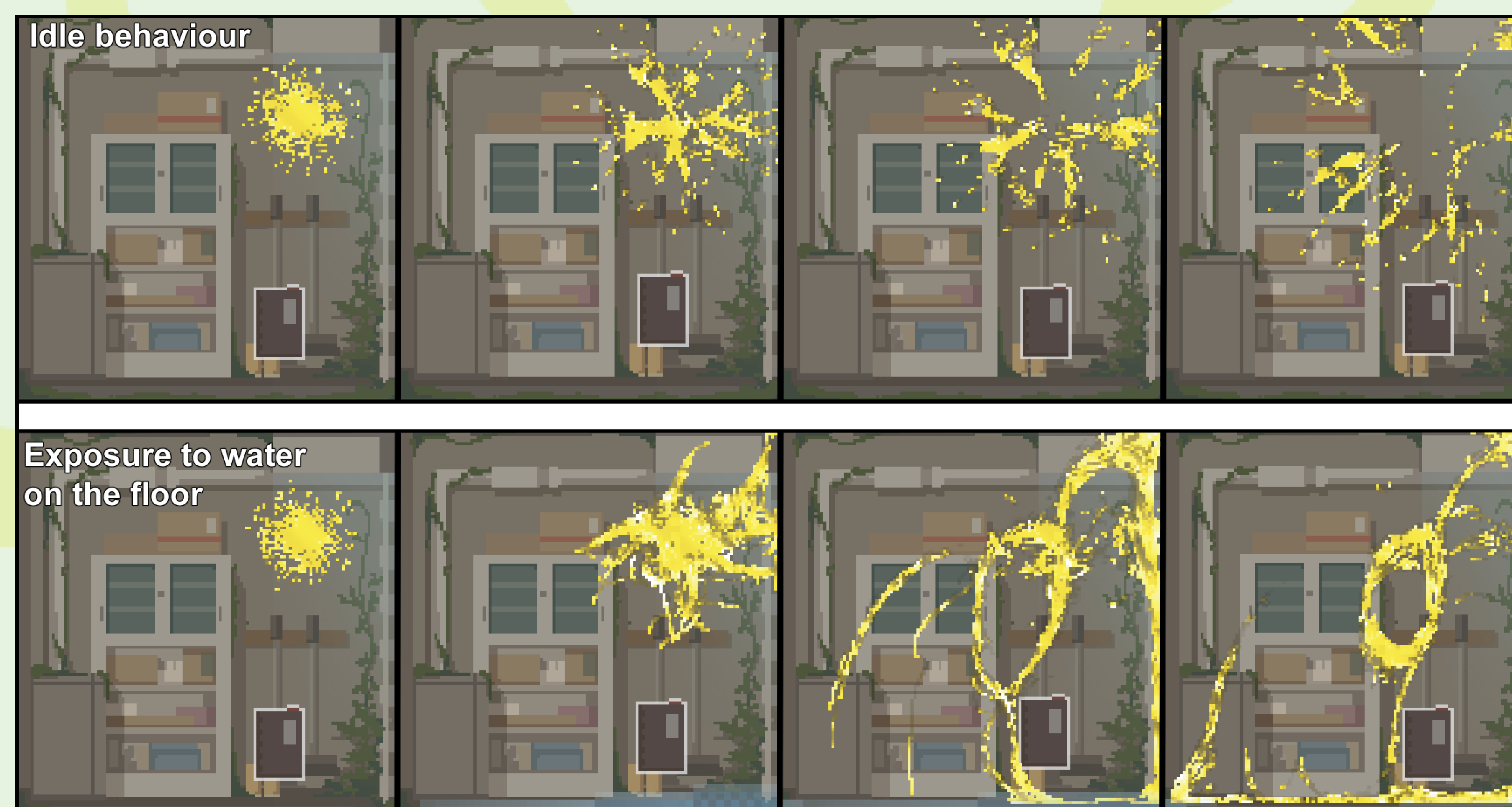
- It is neither plant nor animal.
- It is a non-hierarchical collection of parts, each of a non-dimorphic sexuality, that somehow effortlessly work together as a whole.
- They exhibit pathfinding abilities that rival human intelligence.

We humans have a lot to learn from the slime mold's unique ontology. Through this, we may understand even better than before why nature is so worth protecting.

Our Solution

We have modelled slime molds in software through the usage of the Slime Mold Algorithm (SMA). We have framed this project in the context of a 2D pixel art puzzle game called *Ad Natura*, which takes the player on a short but meaningful journey in an apocalyptic setting, interacting with the slime mold to clear a path forward, and to restore hope to humanity. Each interaction with the slime mold will teach the player something new, and elicit emotions ranging from sadness and fear, to curiosity, empathy, and hope.

EXPERIENCE COLLECTIVE WILL



This visual demonstrates the basic feature of the Slime Mold Algorithm, which is to search for and find paths to food and water sources. It branches out nondeterministically as it searches, and then when it finds a source of sustenance, it reinforces that trail and eliminates obsolete trails.

Tools Used



Unity

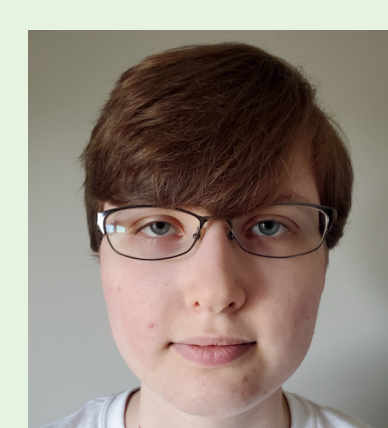


Visual Studio



C-Sharp

Meet the Team



Felix Hurst



Marcos
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Supervisors: Dr. Stephen Kelly, Michelle Bunton

Key Features

Player tools:

- Water Shooter – For helping slime decompose wood
- Impact Rounds – For breaking brittle obstacles
- Wind Fan – For blowing slime spores

Slime mold mechanics:

- Water attraction
- Wood decomposition
- Spore reproduction
- Light aversion
- Death when contacting pollution sources
- Coating walls, making them climbable

